

MRPL | DEMONSTRATION CASE

Mangalore Refinery and Petrochemicals Limited
Independently prepared specimen. Not an MRPL-issued inspection report.

Rotating equipment Condition inspection report

FOR TECHNICAL REVIEW | Status: OPEN | Inspection: 25 September 2026

Inspection record	Asset / assignment
Report: DEMO-MECH-2026-017 Revision: 00 Prepared: 25 Sep 2026	Asset tag: P-102A (fictional) Service: cooling-water circulation (demo)
Discipline: Mechanical / condition monitoring Inspection type: routine review with follow-up	Drive tag: M-102A (fictional) Location: utility pump bay U-01 (fictional)
Prepared by: Demo mechanical inspector Review authority: role placeholder only	Operating state in records: running Live site inspection: none; synthetic dataset

SCOPE AND EVIDENCE

Review eight daily spot readings, the visual inspection sheet and the recorded field checklist. Determine which observations are directly supported, identify records that need reconciliation, and prepare a maintenance-review note with traceable sources.

SUMMARY FOR REVIEW

The recorded vibration and bearing temperature rise over the observation period, while recorded flow falls. The visual attachment and checklist require comparison before the guard and housing condition can be accepted. A small liquid-collection test is recorded only on the scanned attachment. No defect cause, fluid identity or fitness-for-service determination is established.

Boundary: every value and criterion in this specimen is a demonstration input. No actual MRPL equipment, maintenance event, acceptance standard or approved operating limit is represented.

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TABLE M1 - DAILY SPOT READINGS

Synthetic readings at approximately 09:00 local time. Vibration: motor drive-end horizontal overall velocity, mm/s RMS. Temperature: drive-end bearing housing surface. Flow: recorded local indication. No uncertainty budget or calibration certificates are supplied.

Date	Vibration mm/s RMS	Bearing temp. deg C	Flow m3/h
2026-09-18	2.0	54	122
2026-09-19	2.2	55	121
2026-09-20	2.1	54	122
2026-09-21	2.8	56	120
2026-09-22	3.2	59	119
2026-09-23	3.6	62	118
2026-09-24	4.1	64	117
2026-09-25	4.6	67	116

DEMO REVIEW CRITERIA - NOT EQUIPMENT OPERATING LIMITS

Parameter	Flag rule for this exercise
Vibration	Flag a reading strictly greater than 4.0 mm/s RMS.
Bearing temperature	Flag a reading greater than or equal to 65 deg C.
Visual record	Any conflict between a checklist and a visible feature remains open for review.
Calculated trend	Use the first and last dated rows; retain original units and state the formula.

Requested calculation sheet: mean vibration; first-to-last percentage changes in vibration and flow; dates/counts crossing each demo criterion; collected-liquid rate using the scanned field values. Exclude missing values explicitly and reject malformed numeric input. Do not infer a maintenance interval or remaining life from eight observations.

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03 / VISUAL INSPECTION & SCANNED FIELD RECORD

DEMO-MECH-2026-017 | REV 00

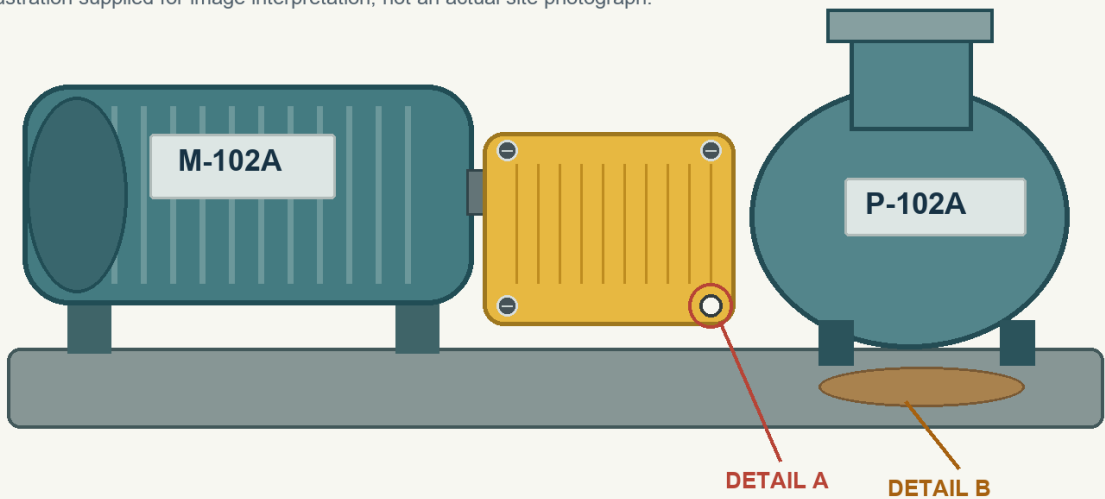
Attachment V1 is intentionally image-only. Read the visible asset tags, compare the image with the marked checklist, and transcribe the liquid-collection entries with their units. Distinguish observations from possible explanations.

FIELD ATTACHMENT V1

P-102A | Synthetic visual evidence | 25 SEP 2026

Inspection view: motor / coupling guard / pump housing

Illustration supplied for image interpretation; not an actual site photograph.



FIELD CHECKLIST | Inspector entry - not yet reconciled

Coupling guard fixing bolts all present: ☒ YES ☐ NO

Collected liquid: 18 mL

Duration: 10 min

Liquid identity: not confirmed

Collection location: pump housing / drain area

Field note:

Check detail A against the ticked checklist.

Residue visible at detail B; origin is unconfirmed.

Inspection status: OPEN - reviewer assessment pending

Recorded by: DEMO INSPECTOR Endorsement: not signed

SYNTHETIC SAMPLE - NOT ISSUED OR APPROVED BY MRPL

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04 / DISPOSITION & REVIEW RECORD

DEMO-MECH-2026-017 | REV 00

OPEN ACTION REGISTER

ID	Review action / evidence	Owner / status
A-01	Reconcile the guard checklist with visible detail A on V1. Describe what is visible and what remains uncertain.	Mechanical reviewer OPEN
A-02	Review detail B and the collection entries. Record the calculated rate separately from any conclusion about fluid identity or source.	Inspection reviewer OPEN
A-03	Validate calculations against Table M1. Report threshold-crossing dates and units using the demo criteria on page 2.	Condition monitoring OPEN
A-04	Draft a review note separating observations, calculations, limitations and proposed follow-up. Link each item to M1 or V1.	Maintenance planner OPEN

INFORMATION REQUIRED BEFORE ANY REAL-WORLD DISPOSITION

Applicable equipment limits; measurement-method and calibration records; operating duty and load at each reading; repeat measurements; confirmation of the visually observed condition; approved inspection and maintenance procedures. These are absent from the demonstration package.

DOCUMENT CONTROL AND ENDORSEMENT

Role	Review record
Prepared by	Demo mechanical inspector synthetic specimen not signed
Checked by	Mechanical reviewer pending no real person identified
Approved by	Authorized organization representative NOT APPROVED

Evidence register: M1 - eight-row synthetic measurement record (page 2 and companion CSV); V1 - illustrative equipment view and scanned field checklist (page 3 and companion PNG). The same report ID appears throughout for traceability.

Use restriction: training, UI demonstrations and local-model evaluation only. This specimen does not represent MRPL's actual document format, policies, assets or an approved inspection record. No logo, seal or real signature has been reproduced.